Reaction-wheel comparison for spiral-outward assembly

Goal: hold structure attitude at zero (exact cancellation). The wheel must store h_w = -h_rel and apply torque -dh_rel/dt.

What the wheels must do at full robot speed (analytic, 0.01 s grid)

Quantity	x	y	z	Unit
Momentum, peak per axis (x / y / z)	1.09	1.14	7.33	N m s
Torque, peak per axis (x / y / z)	11.18	11.64	17.99	N m
Peak momentum, vector norm	7.33	at t = 283 s		N m s
Peak torque, vector norm	17.99	at t = 283 s		N m
Final wheel momentum	-0.0e+00	2.2e-18	4.3e-17	N m s
Nominal robot motion time	309 s	5.15 min		

Headline

- Full-speed need is far above any flown wheel's torque. Fix: slow each robot move until the wheel keeps up (cost: assembly time).
- Torque, not momentum, limits the mission. RW8 (more momentum) gives the same time as RW4 (same 0.25 N m).
- 4RW0 (CubeSat unit) needs 290 min of motion per-move (1508 min uniform).
- BCT RW4-HT x3 (recommended in ZO-25): 13.5 min per-move, 34.5 min uniform, 9.6 kg.
- Open: wheel power at full speed (5.1 kW real peak) is why slowing is needed; wheel mass (up to 9.6 kg) is not yet in the inertia.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Candidate wheel sets (vendor datasheet values)

Per-axis limits. BCT sets are three wheels, one per body axis. 4RW0 is a 4-wheel pyramid; per-axis values are as given on its product page.

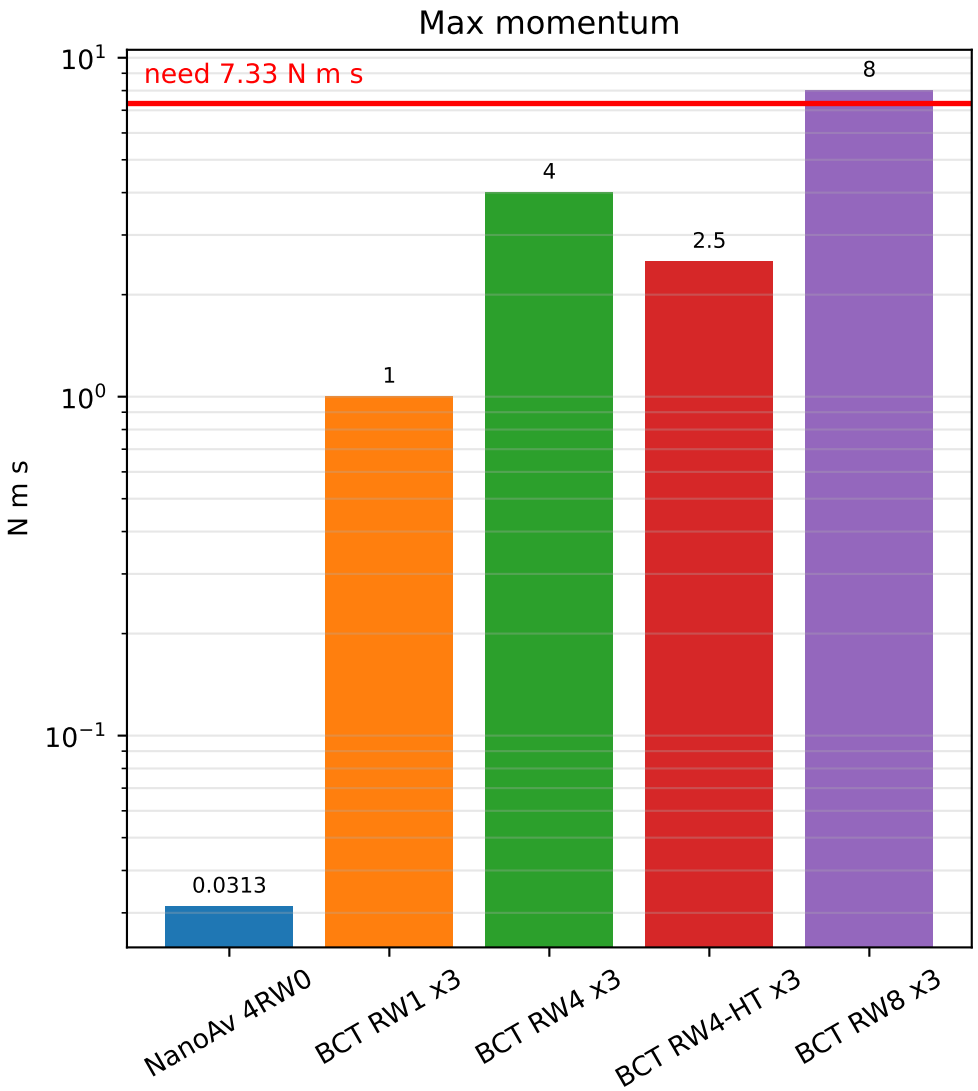
Wheel set	H max x/y/z (N m s)	T max x/y/z (N m)	Mass (kg)	Peak power (W, total)	Heritage
NanoAv 4RW0	0.037/0.037/0.0313	0.0059/0.0059/0.005	0.67	n/a	CubeSat-class unit (vendor page; flight count not stated in screenshots)
BCT RW1 x3	1/1/1	0.06/0.06/0.06	3.30	300	RW1: 162 units launched, 208 cumulative years, longest 3.3 years
BCT RW4 x3	4/4/4	0.25/0.25/0.25	9.60	528	RW4 family: 94 units launched, 190 cumulative years on orbit, longest unit 5.2 years
BCT RW4-HT x3	2.5/2.5/2.5	0.5/0.5/0.5	9.60	528	RW4 family: 94 units launched, 190 cumulative years on orbit, longest unit 5.2 years
BCT RW8 x3	8/8/8	0.25/0.25/0.25	13.20	528	RW8 family: 14 units launched, 44 cumulative years, longest 5.2 years
NEED at full speed	1.09/1.14/7.33	11.18/11.64/17.99	-	5100 (real peak)	ZO-25 analytic requirement

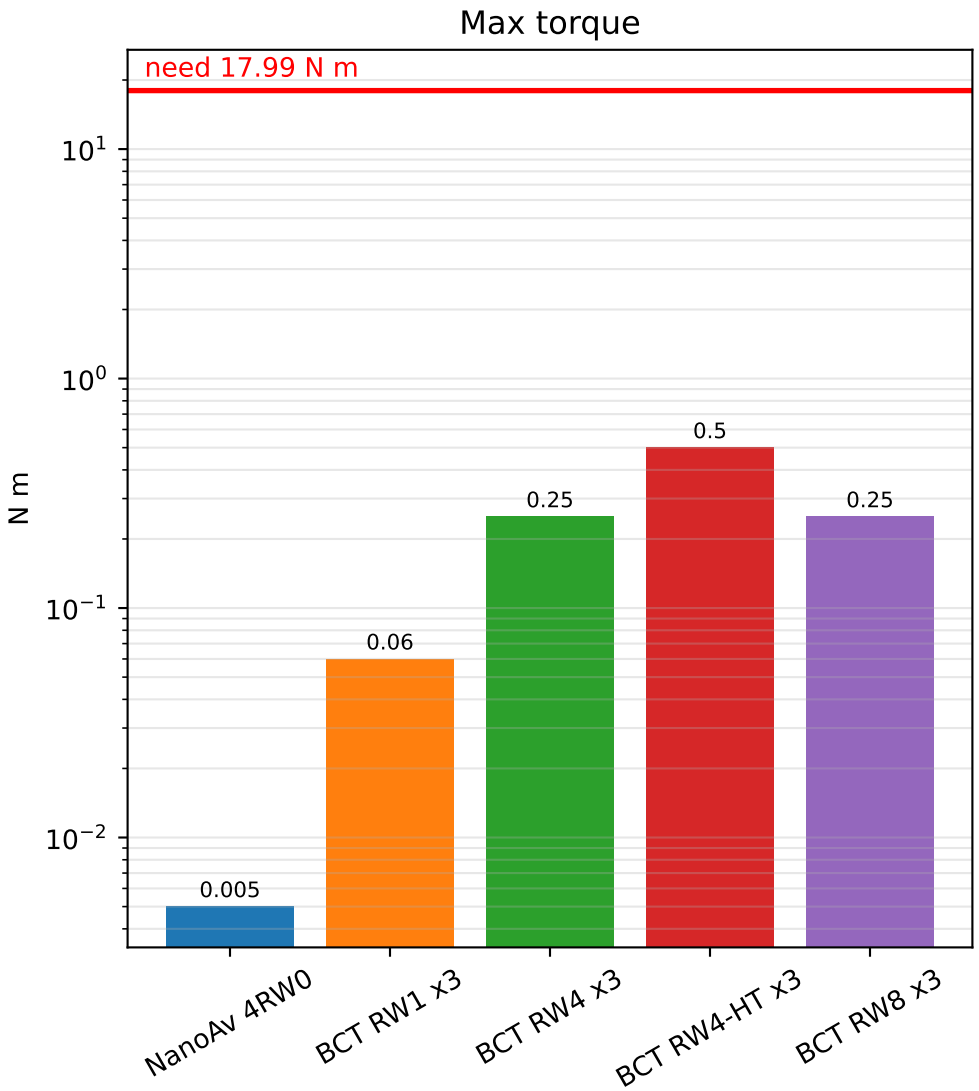
Sources: NanoAvionics 4RW0 product page (user screenshots in 'Reaction Wheel/'); Blue Canyon Technologies, 'Reaction Wheels Full Product Line', REV 1/2026. RW4-HT trades momentum (2.5 vs 4.0 N m s) for double torque (0.50 vs 0.25 N m). BCT heritage numbers are for the RW4 family; the datasheet does not split RW4 and RW4-HT.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Capability against need (z axis, the worst axis)

Bars are wheel limits. Red line is the full-speed need. Anything below the red line forces the robot to slow down.



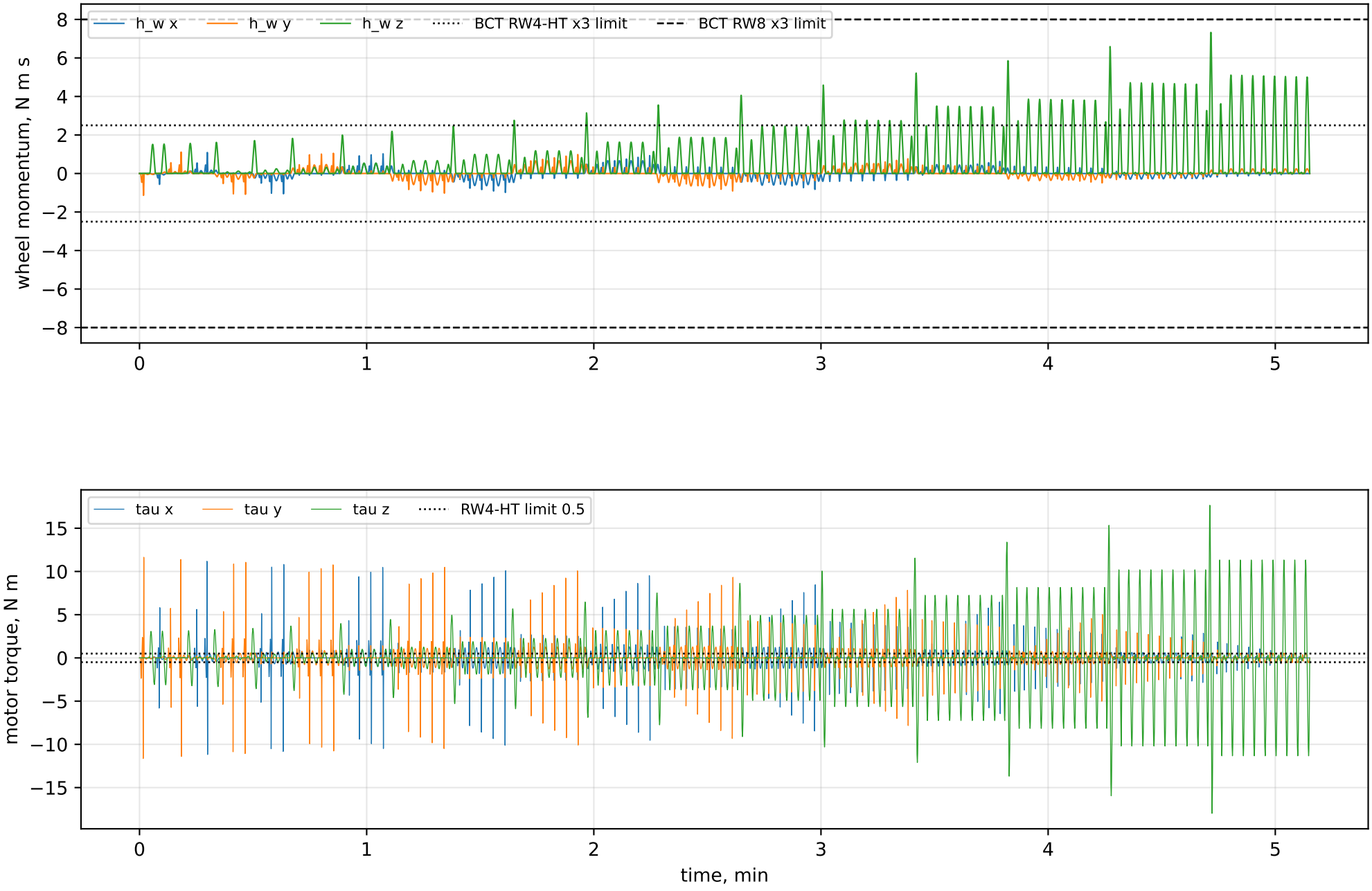


Torque shortfall is the larger gap: even the best set (0.50 N m) is ~36x below the 17.99 N m peak. Momentum shortfall is ~3x for RW4-HT and absent only for RW8 (8 N m s > 7.33). Slowing a move by s cuts momentum need by s and torque need by s^2.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

What the spiral assembly asks of the wheels (full speed)

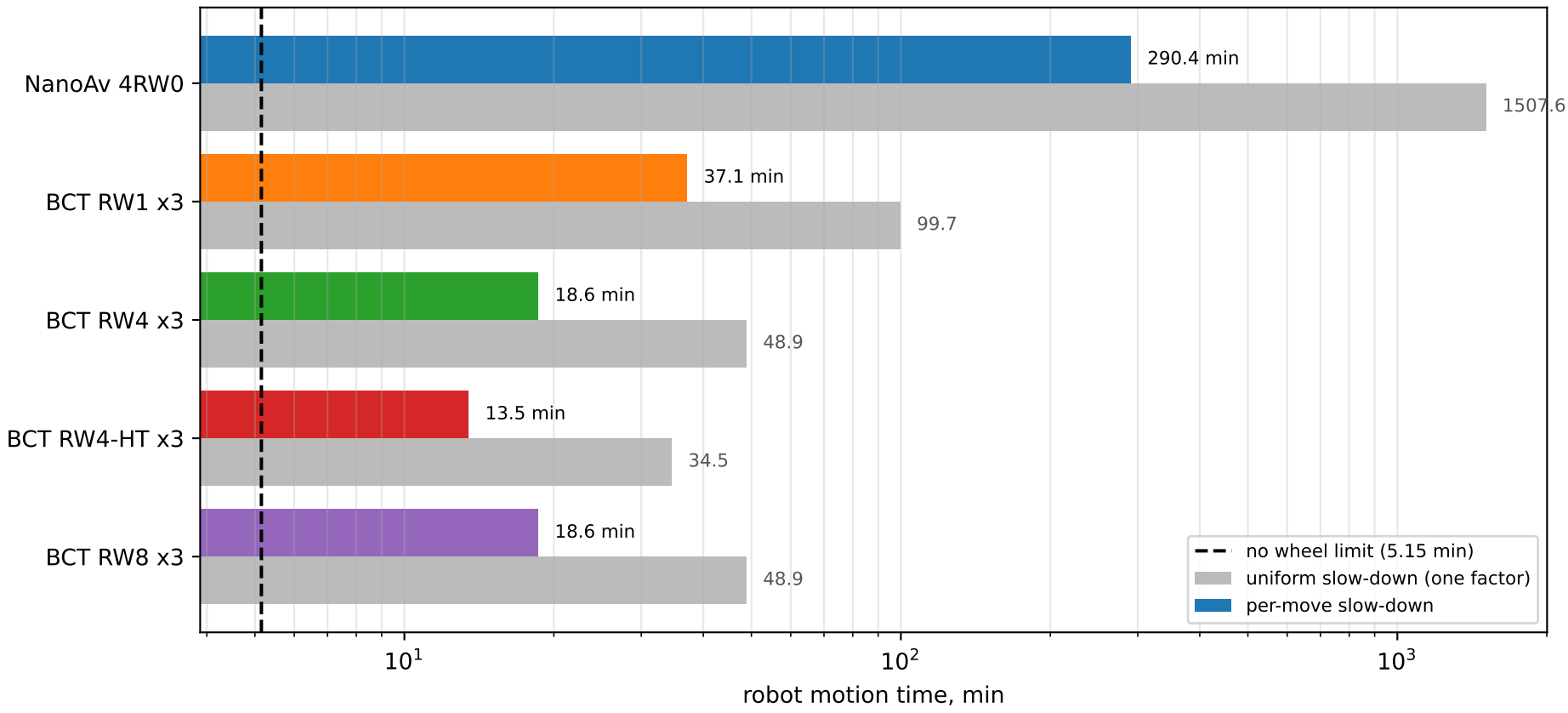
Wheel momentum h_w = -h_rel and motor torque -dh_rel/dt, per body axis. Dotted lines: limits of two candidate sets.



Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Cost of using each wheel: robot motion time

Robot moves are slowed (same path) until the wheel stays within 80% of its momentum and torque limits. Log scale.

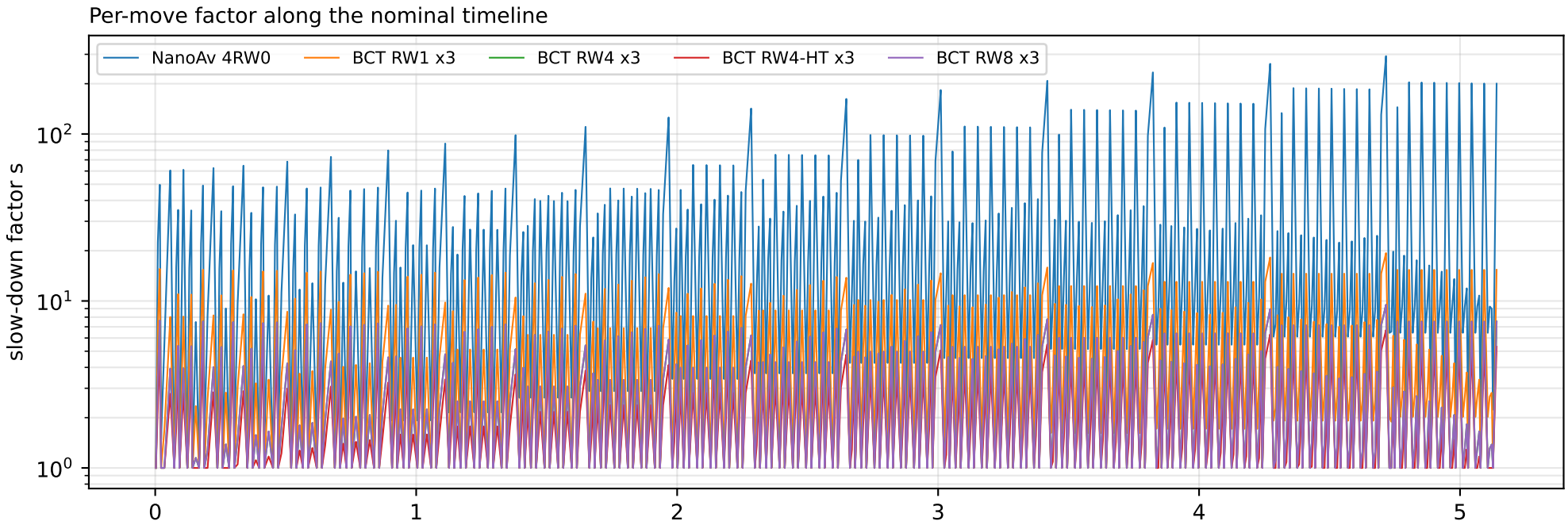


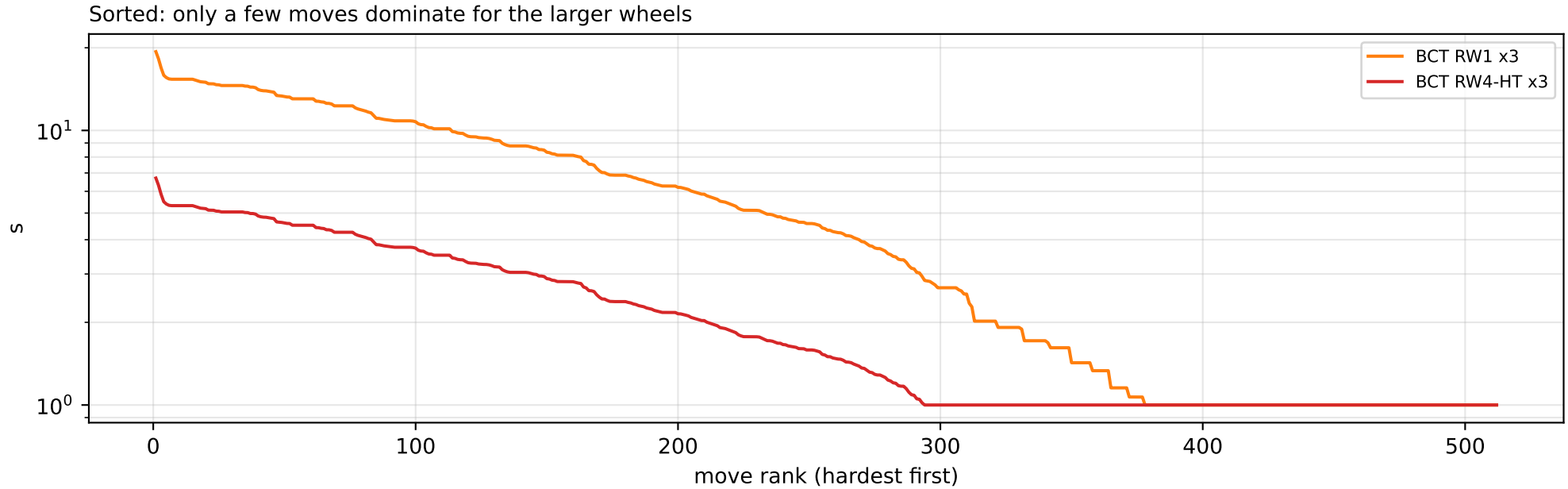
Wheel set	Per-move (min)	Uniform (min)	Per-move vs nominal	Worst move slow-down s	Moves limited by torque / momentum / no	Mass (kg)
NanoAv 4RW0	290.4	1507.6	56.4x	292.7	396 / 102 / 14	0.7
BCT RW1 x3	37.1	99.7	7.2x	19.4	377 / 0 / 135	3.3
BCT RW4 x3	18.6	48.9	3.6x	9.5	312 / 0 / 200	9.6
BCT RW4-HT x3	13.5	34.5	2.6x	6.7	293 / 0 / 219	9.6
BCT RW8 x3	18.6	48.9	3.6x	9.5	312 / 0 / 200	13.2

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Where the time goes: slow-down factor for each of the robot moves

512 moves between rest states. Factor s = 1 means the move runs at nominal speed.

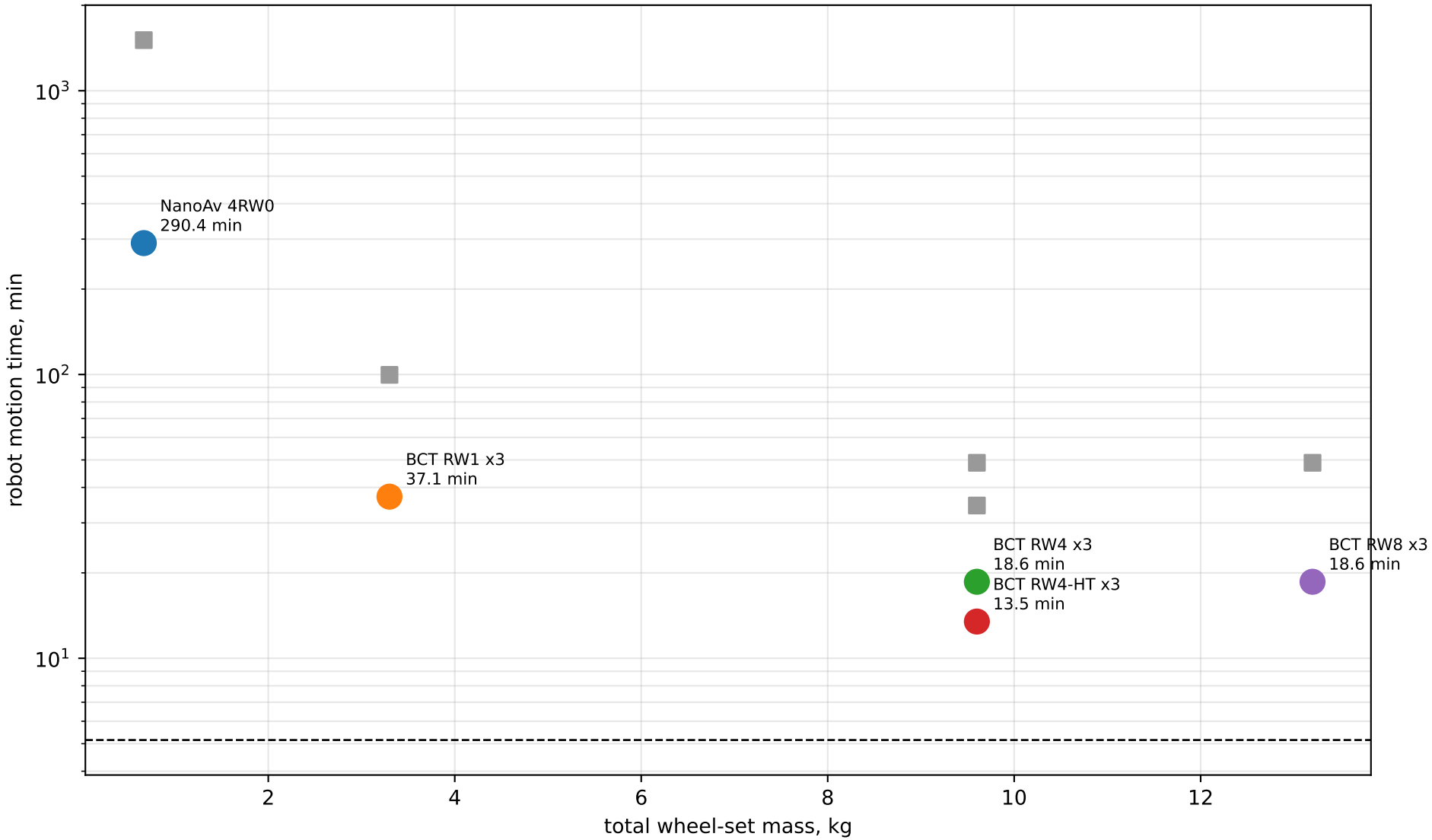




Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Trade: motion time against wheel mass

Marker label = per-move motion time. Lower left is better. Uniform times are the grey markers.

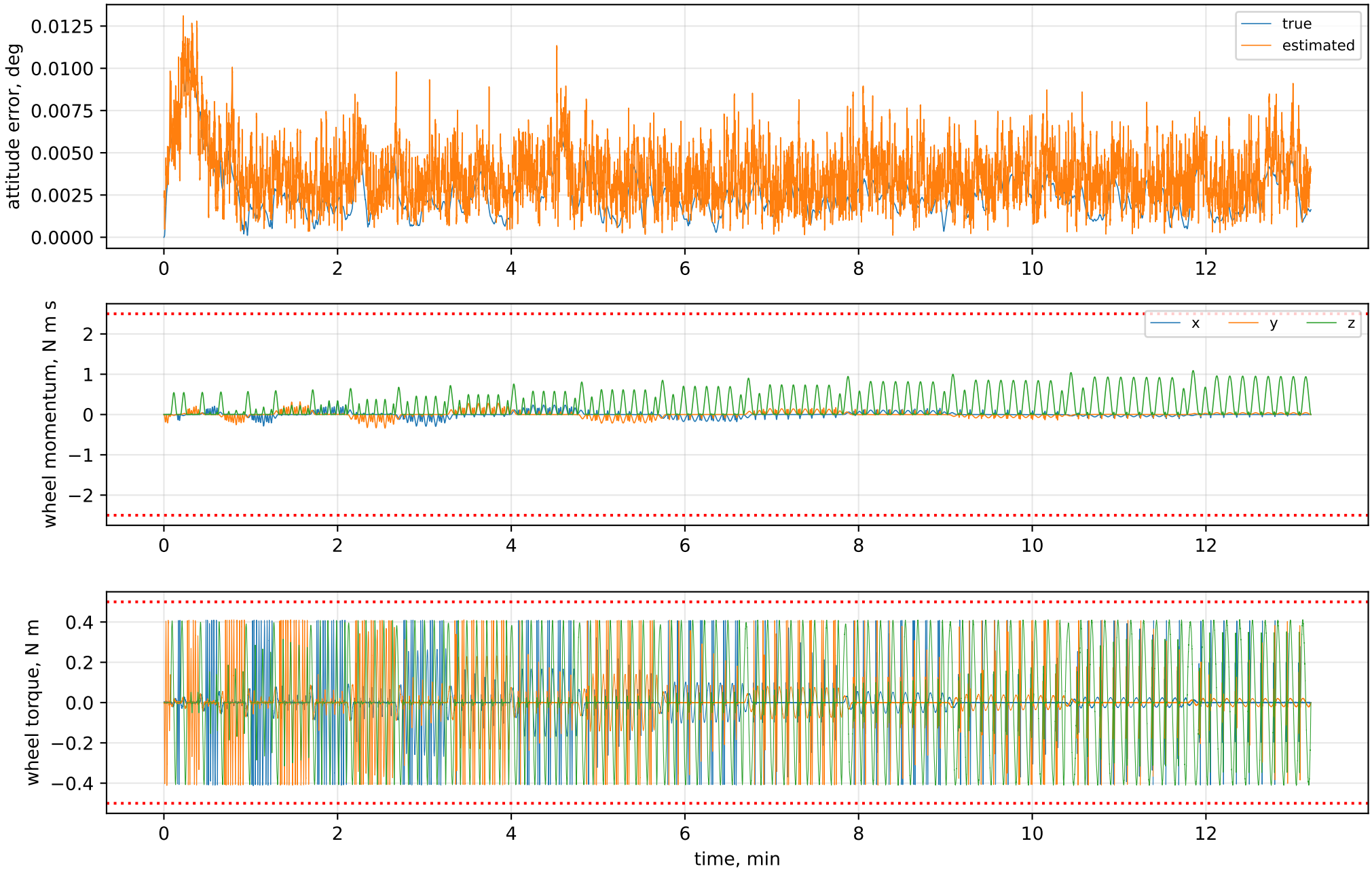


Heavier wheels buy back time, but the buy stops when torque saturates (RW8 = RW4 in time, 1.2 kg heavier per axis).
Wheel power (176 W peak per BCT RW4/RW8 wheel; 100 W RW1) must come from the structure power system.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Closed-loop check: BCT RW4-HT x3 (high torque)

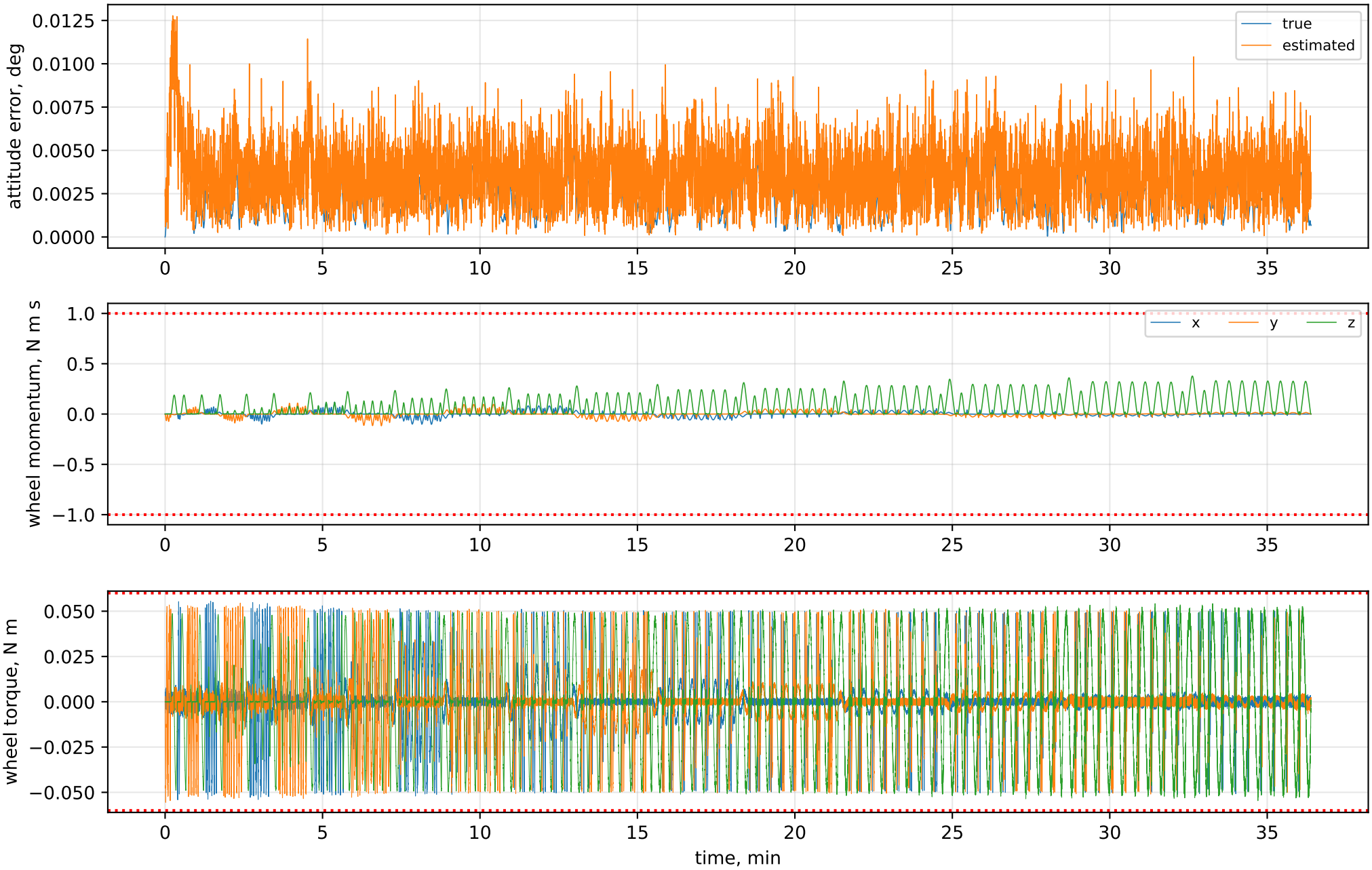
Flight-style controller, noisy gyro and star tracker, 1-cycle report delay, robot 2% faster than plan. Motion 13.5 min. Peak true attitude error 0.0103 deg, final 0.0016 deg. Limit hits: {'sat_torque': 0, 'sat_momentum': 0}.



Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Closed-loop check: BCT RW1 x3

Flight-style controller, noisy gyro and star tracker, 1-cycle report delay, robot 2% faster than plan. Motion 37.1 min. Peak true attitude error 0.0102 deg, final 0.0007 deg. Limit hits: {'sat_torque': 0, 'sat_momentum': 0}.



Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).